

Progress Report: Sprint 6

CS.028 Optimizing Research Software Codes - 5 January 2026

[Kathryn Butler](#), [Michael McAllister](#), [Joseph Schaab](#)

Summary

- Continued testing performance, and found a confusing bug with timing issues.
- The biggest blocker is limited team availability with changing schedules.
- Smaller datasets are needed for more consistent timing.

Sprint Work Log

Anomalous Timing

Status: Partial

Author: Kathryn Butler

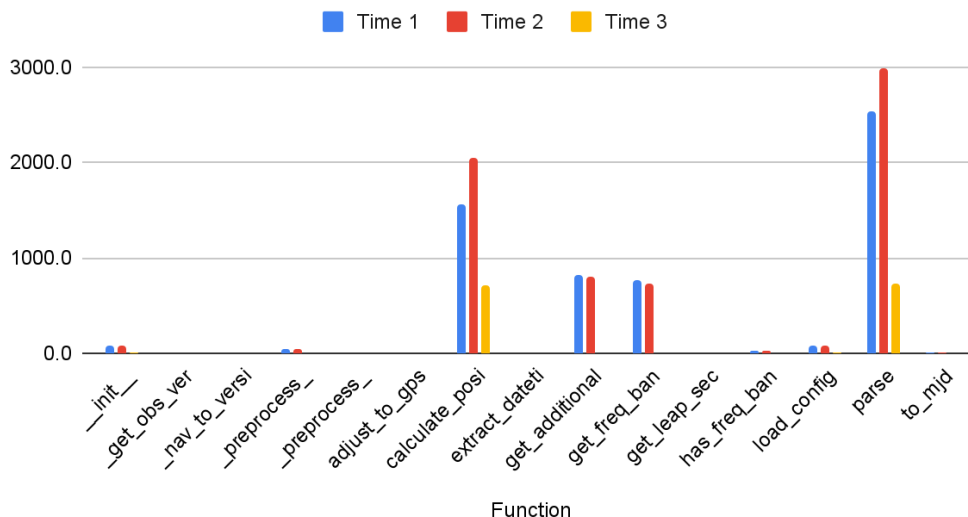
Reviewers: N/A

Source Link:

<https://github.com/KathrynJButler/optimizing-research-software-codes/issues/4>

Supporting Artifact:

Time 1, Time 2 and Time 3



Notes: Confusing function timing with the same data. I need to research more on why this is happening, and how to solve it. Also add another sample data file which is shorter and can provide a faster runtime for testing. It is also difficult to set up the environment consistently. Look into updating the documentation to include a more detailed setup process.

Planning Long Term Goals

Status: Complete

Author: Joseph Schaab

Reviewers: N/A

Source Link:

https://oregonstateuniversity-my.sharepoint.com/:o:/g/personal/schaabj_oregonstate_edu/IgDuuo4Ss4XuTLPRII_mTSy7AbzS2ymWA3N3VTg2ks-pntQ

Supporting Artifact: In our project OneNote folder, under Project Partner Meetings Notes, Winter Term Goals has been added to establish what we need to complete by the end of the term.

Winter Term Goals

Sunday, January 18, 2026 11:32 PM

- Add detailed commenting to the main source code file
- Draft efficiency improvements for function calculate_posi()
- Draft efficiency improvements for function parse()
- Test calculate_posi() with efficiency improvements implemented
- Test parse() with efficiency improvements implemented
- Improve the buggy runtime inefficiencies
- Provide a consistent and optimized program by end of term

Notes: Met with project partner on Jan 15, 2026. Discussed our long term goals to work towards this term. This list notes what we need to complete in order to stay on track by the start of spring term. We will update these as needed and discuss with our project partner before altering any of these goals.

Risk and Quality Tracking

Bug Count: 2 ([Link to issue query](#))

Top Bugs:

- [Inconsistent timing data](#)
 - **Severity:** High
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Create a smaller test data pool, run the tests multiple times to determine if the size of the file increases the likelihood of failed parsing attempts.

- [Incorrect conversion from float to datetime](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Research into the line causing the problem. Determine the input causing the problem, and correct the datetime variable.

Next Sprint Plan

- Create a smaller sample test dataset for faster testing.
- Continue researching into timing inconsistencies and csv parsing failures.
- Update documentation to improve the setup process.

Team Process Reflection

Our team scheduled times to meet with our project partner and TA. We also ran more tests on the sample database, and found a strange timing inconsistency that seems to also fail at parsing the csv file.

The main thing our team needs to focus on is communication with both the TA and Dr. Park, as well as inter-team communication. The majority of the sprint was spent attempting to find a scheduled meeting time between all of our schedules, detracting from improvements to the code.

Actionable improvements we will do is create more consistent updates throughout the term to both our TA and Dr. Park. This will be evidenced by updates in the progress-updates channel in our Discord group with the TA, and email reports to Dr. Park. We will also communicate with each other more frequently.